CLIMATE ACTION RESERVE

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Variance Determination

Project Name:	Indigo U.S. Project No. 1
Project ID Number:	CAR1459
Project Developer:	Indigo Ag.
Protocol:	Soil Enrichment Protocol v1.1
Date:	6/14/23

The Climate Action Reserve (Reserve) received a Request for Project Variance from Indigo Ag. on 5/25/2023 in regards to CAR1459, the Indigo U.S. Project No. 1 (U.S. Project No. 1). U.S. Project No. 1 has been submitted to the Reserve under version 1.1 of the Soil Enrichment Protocol and is subject to all requirements of the protocol.

The protocol requires that "GHG emission reductions from a soil enrichment project are quantified by comparing modeled and calculated project and baseline emissions, as well as from modeled/measured changes in SOC."

Indigo Ag. has requested a variance for U.S. Project No. 1 to estimate grazing impacts on CO_2 emissions from the carbon pool using a combination of approved SEP-derived default equations, and literature-based estimates, as there are not sufficient published datasets yet available for grazing impacts on soil organic carbon in cropland systems available to meet the model calibration and validation requirements for DayCent-CR.

The Reserve grants this variance pursuant to the following conditions:

- Indigo Ag. utilizes the parameters and conditions described in the attached "Proposal to estimate grazing impacts on CO_2 emissions using a simple, conservative approach to allow crediting of grazed fields" document, including the requirements that:
  - Indigo Ag. estimates the reduction in carbon inputs to the soil due to removal of plant biomass as forage using the following equation:
    - Equation 1. Carbon dioxide losses due to grazing activities not included in SOC modeling:

$$CO2_{grazing_SOC} = \Sigma\ (AGD_i \times TAM_i \times DMI \times Frac_C \times DE_i \times Frac_{stabilization} \times 44/12 \times 1/1000$$

    - Indigo Ag. Uses the conservative $Frac_{stabilitzation}$ value of 0.25.
- The verifier confirms that the conditions contained in the "Proposal to estimate grazing impacts on CO_2 emissions using a simple, conservative approach to allow crediting of grazed fields" are reasonable and conservative.
- The project developer agrees that any estimates derived using this proposal will be trued-up at the 5 year deadline using direct measurements and that CRT issuance is adjusted as needed as a result.
- The project developer agrees that future modeling shall be used to estimate grazing impacts on CO_2 emissions as it becomes available, and that CRT issuance may be adjusted as a result.
- The project developer agrees that any changes to the proposed equation with a 2.0 update must be retroactively applied/corrected to the affected reporting period, and that CRT issuance may be adjusted as a result.
- The verifier confirms all other relevant quantification and modeling requirements were met.

This variance is expressly limited to U.S. Project No. 1 as specified above for the period from 1/1/2022 - 12/31/2022. Indigo Ag. must comply with all other requirements of the protocol. Further, this variance does not set precedent for future reporting periods, projects, or Reserve decisions.

Signed:



Chloe Ney
Analytical Associate
Climate Action Reserve